OGC® DOCUMENT:
INSITU_OBSERVATIONS_PROFILE

Open Geospatial Consortium

IN-SITU OBSERVATIONS EDR PROFILE

DRAFT STANDARD
Implementation

APPROVED

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ABSTRACT

This document defines the In-situ Observations EDR Profile, an OGC API — Environmental Data Retrieval (EDR) Part 3 Service Profile. It specifies normative requirements and conformance tests for server implementations conforming to this profile.

II PREFACE

This document was prepared by the submitting organizations.

III SECURITY CONSIDERATIONS

No security considerations have been made for this document.

IV SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table — Submitters

NAME	AFFILIATION
Unknown *(editor)*	

1 SCOPE

1 SCOPE

This standard defines the In-situ Observations EDR Profile. It specifies requirements and conformance tests for implementations of this profile.

2

CONFORMANCE

2 CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/insitu_observations_profile

3 NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

4 TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

5 REQUIREMENTS

5 REQUIREMENTS

NOTE: Conformance with this profile is assessed against its requirements ("shall") only. Recommendations ("should") and permissions ("may") are provided as informative guidance: they are not conformance-tested and an implementation is not made non-conformant by not following them. See the quality-assessment summary emitted by the profile tooling for a count of the recommendations and permissions defined by this profile.

REQUIREMENTS CLASS 1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
CONFORMANCE CLASS	Conformance class A.1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/insitu_observations_profile`
TARGET-TYPE	In-situ Observations EDR Profile Profile Standard
NORMATIVE STATEMENTS	Requirement 1: `/req/insitu_observations_profile/conformance` Requirement 2: `/req/insitu_observations_profile/collections-endpoint` Requirement 3: `/req/insitu_observations_profile/collection-crs` Requirement 4: `/req/insitu_observations_profile/collection-temporal-extent` Requirement 5: `/req/insitu_observations_profile/collection-custom-dimensions` Requirement 6: `/req/insitu_observations_profile/data-queries` Requirement 7: `/req/insitu_observations_profile/output-formats` Requirement 8: `/req/insitu_observations_profile/parameter-names` Requirement 9: `/req/insitu_observations_profile/locations-response` Requirement 10: `/req/insitu_observations_profile/error-handling`

REQUIREMENT 1	
IDENTIFIER	`/req/insitu_observations_profile/conformance`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The service SHALL declare conformance to OGC API EDR Core and the MetOcean EDR Profile.
A	The /conformance endpoint SHALL include http://www.opengis.net/spec/ogcapi-edr-1/1.0/conf/core.
B	The /conformance endpoint SHALL include https://eumetnet.github.io/spec/metocean-edr-profile/1/conf/core.

REQUIREMENT 1	
C	The /conformance endpoint SHALL include https://eumetnet.github.io/spec/metocean-edr-profile/1/conf/insitu-observations.

REQUIREMENT 2	
IDENTIFIER	`/req/insitu_observations_profile/collections-endpoint`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The service SHALL provide a /collections endpoint listing the insitu-observations collection.
A	The /collections endpoint SHALL return the insitu-observations collection.
B	Each collection SHALL include extent, data_queries, output_formats, and parameter_names.

REQUIREMENT 3	
IDENTIFIER	`/req/insitu_observations_profile/collection-crs`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The insitu-observations collection SHALL support OGC CRS84 as its primary CRS.
A	The extent.spatial.crs SHALL be http://www.opengis.net/def/crs/OGC/1.3/CRS84.
B	The crs array SHALL include http://www.opengis.net/def/crs/OGC/1.3/CRS84.

REQUIREMENT 4	
IDENTIFIER	`/req/insitu_observations_profile/collection-temporal-extent`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The insitu-observations collection SHALL declare a temporal extent using the Gregorian TRS.
A	The extent.temporal.trs SHALL be Gregorian.

REQUIREMENT 4

B	The extent.temporal.interval SHALL contain at least one valid ISO 8601 interval.

REQUIREMENT 5

IDENTIFIER	`/req/insitu_observations_profile/collection-custom-dimensions`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The insitu-observations collection SHALL define standard_name and level custom dimensions.
A	extent.custom SHALL contain an object with id standard_name and reference https://vocab.nerc.ac.uk/standard_name/.
B	extent.custom SHALL contain an object with id level and reference describing measurement height in meters.

REQUIREMENT 6

IDENTIFIER	`/req/insitu_observations_profile/data-queries`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The insitu-observations collection SHALL support locations, area, and radius data queries.
A	The collection SHALL provide a /collections/insitu-observations/locations endpoint.
B	The collection SHALL provide a /collections/insitu-observations/area endpoint.
C	The collection SHALL provide a /collections/insitu-observations/radius endpoint.
D	The radius query variables SHALL include within_units containing at least m.

REQUIREMENT 7

IDENTIFIER	`/req/insitu_observations_profile/output-formats`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`

REQUIREMENT 7

STATEMENT	All data queries SHALL support CoverageJSON as a response format.
A	All data queries SHALL support application/vnd.cov+json.
B	The locations query SHALL additionally support application/geo+json.

REQUIREMENT 8

IDENTIFIER	`/req/insitu_observations_profile/parameter-names`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	Each parameter in parameter_names SHALL conform to the MetOcean EDR Profile parameter conventions.
A	Each parameter SHALL specify label, description, unit, observedProperty, and measurementType.
B	unit.symbol.type SHALL be a QUDT unit URI of the form https://qudt.org/vocab/unit/<unit>.
C	observedProperty.id SHALL be a CF standard name URI of the form https://vocab.nerc.ac.uk/standard_name/<name>.
D	Each parameter SHALL specify metocean:standard_name matching a value in extent.custom standard_name.
E	Each parameter SHALL specify metocean:level matching a value in extent.custom level.

REQUIREMENT 9

IDENTIFIER	`/req/insitu_observations_profile/locations-response`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The locations query SHALL return a GeoJSON FeatureCollection of station locations.
A	The response Content-Type SHALL be application/geo+json.
B	Each feature SHALL have an id property of type string.
C	Each feature SHALL have a properties.name property of type string.

REQUIREMENT 10

IDENTIFIER	`/req/insitu_observations_profile/error-handling`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
STATEMENT	The service SHALL handle errors per RFC 9457.
A	A 4xx response SHALL include a body conforming to RFC 9457 Problem Details.
B	A query for data inside the extent that returns no data SHALL respond with HTTP 204.

A

ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

A ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

CONFORMANCE CLASS A.1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/insitu_observations_profile`
REQUIREMENTS CLASS	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/insitu_observations_profile`
CONFORMANCE TESTS	Abstract test A.1-1: `/conf/insitu_observations_profile/conformance` Abstract test A.1-2: `/conf/insitu_observations_profile/collections-endpoint` Abstract test A.1-3: `/conf/insitu_observations_profile/collection-crs` Abstract test A.1-4: `/conf/insitu_observations_profile/collection-temporal-extent` Abstract test A.1-5: `/conf/insitu_observations_profile/collection-custom-dimensions` Abstract test A.1-6: `/conf/insitu_observations_profile/data-queries` Abstract test A.1-7: `/conf/insitu_observations_profile/output-formats` Abstract test A.1-8: `/conf/insitu_observations_profile/parameter-names` Abstract test A.1-9: `/conf/insitu_observations_profile/locations-response` Abstract test A.1-10: `/conf/insitu_observations_profile/error-handling`

Example 1

label	/conf/insitu_observations_profile/conformance
subject	/req/insitu_observations_profile/conformance
test-purpose	Validate that conformance is correctly implemented.

Send GET request to /conformance.

Verify HTTP status is 200.

Verify conformsTo includes http://www.opengis.net/spec/ogcapi-edr-1/1.0/conf/core.

Verify conformsTo includes https://eumetnet.github.io/spec/metocean-edr-profile/1/conf/core.

Verify conformsTo includes https://eumetnet.github.io/spec/metocean-edr-profile/1/conf/insitu-observations.

Example 2

label	/conf/insitu_observations_profile/collections-endpoint

subject	/req/insitu_observations_profile/collections-endpoint
test-purpose	Validate that collections endpoint is correctly implemented.

Send GET request to /collections.

Verify HTTP status is 200.

Verify the collections array contains an object with id insitu-observations.

Verify the collection includes extent, data_queries, output_formats, and parameter_names.

Example 3

label	/conf/insitu_observations_profile/collection-crs
subject	/req/insitu_observations_profile/collection-crs
test-purpose	Validate that collection crs is correctly implemented.

Send GET request to /collections/insitu-observations.

Verify extent.spatial.crs equals http://www.opengis.net/def/crs/OGC/1.3/CRS84.

Verify the crs array includes http://www.opengis.net/def/crs/OGC/1.3/CRS84.

Example 4

label	/conf/insitu_observations_profile/collection-temporal-extent
subject	/req/insitu_observations_profile/collection-temporal-extent
test-purpose	Validate that collection temporal extent is correctly implemented.

Send GET request to /collections/insitu-observations.

Verify extent.temporal.trs equals Gregorian.

Verify extent.temporal.interval contains at least one two-element array.

Example 5

label	/conf/insitu_observations_profile/collection-custom-dimensions
subject	/req/insitu_observations_profile/collection-custom-dimensions
test-purpose	Validate that collection custom dimensions is correctly implemented.

Send GET request to /collections/insitu-observations.

Verify extent.custom contains an object with id standard_name.

Verify that object has reference https://vocab.nerc.ac.uk/standard_name/.

Verify extent.custom contains an object with id level.

Example 6

label	/conf/insitu_observations_profile/data-queries
subject	/req/insitu_observations_profile/data-queries
test-purpose	Validate that data queries is correctly implemented.

Send GET request to /collections/insitu-observations.

Verify data_queries contains locations, area, and radius.

Verify the radius variables.within_units array contains m.

Example 7

label	/conf/insitu_observations_profile/output-formats
subject	/req/insitu_observations_profile/output-formats
test-purpose	Validate that output formats is correctly implemented.

Send GET request to /collections/insitu-observations/area with a valid polygon.

Verify HTTP status is 200.

Verify Content-Type is application/vnd.cov+json.

Send GET request to /collections/insitu-observations/locations.

Verify Content-Type is application/geo+json.

Example 8

label	/conf/insitu_observations_profile/parameter-names
subject	/req/insitu_observations_profile/parameter-names
test-purpose	Validate that parameter names is correctly implemented.

Send GET request to /collections/insitu-observations.

For each parameter in parameter_names, verify label, description, unit, observedProperty, and measurementType are present.

Verify unit.symbol.type starts with https://qudt.org/vocab/unit/.

Verify observedProperty.id starts with https://vocab.nerc.ac.uk/standard_name/.

Verify metocean:standard_name is present and matches a value in extent.custom standard_name.values.

Verify metocean:level is present and matches a value in extent.custom level.values.

Example 9

label	/conf/insitu_observations_profile/locations-response
subject	/req/insitu_observations_profile/locations-response

test-purpose	Validate that locations response is correctly implemented.

Send GET request to /collections/insitu-observations/locations.

Verify HTTP status is 200.

Verify Content-Type is application/geo+json.

Verify response type is FeatureCollection.

Verify each feature has id and properties.name.

Example 10

label	/conf/insitu_observations_profile/error-handling
subject	/req/insitu_observations_profile/error-handling
test-purpose	Validate that error handling is correctly implemented.

Send GET request to /collections/nonexistent-collection.

Verify HTTP status is 404.

Verify Content-Type is application/problem+json.

Verify response body includes type and title properties.